Simulating Ant Swarm Aggregations Dynamics

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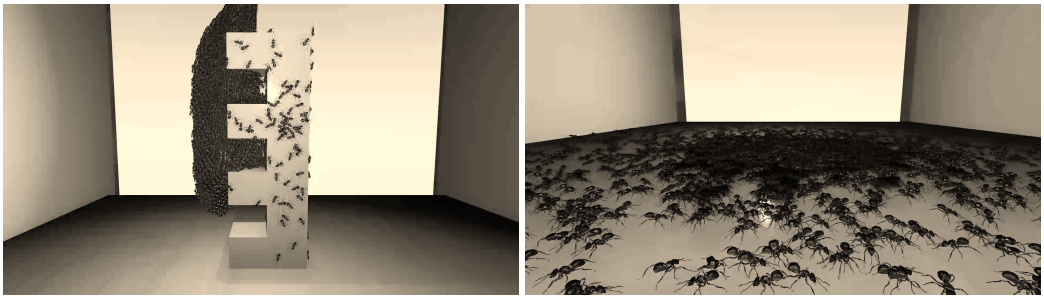


Fig. 1. Left: a flow of ants coming from the top touches on a bracket. The majority of ants move together in a viscous, fluid-like manner, while some ants break away to explore the experimental area. Right: scattering of an ant swarm over a confined space.

Ants exhibit unique abilities to self-assemble into animate, living structures. Such structures display properties of both fluid and solid-like, deformable materials. Despite much progress in our understanding of ant aggregation dynamics, simulating such phenomena has been largely overlooked in real-time graphics and animation applications. We present a constraints-based approach for simulating the collective dynamics of ants interactively with compelling physical realism. Through several experiments and comparisons, we demonstrate that our method can efficiently capture the underlying physical rules of such aggregations, which can inform further research into living structures that can morph and self-repair.

CCS Concepts: • **Computing methodologies** → **Physical simulation**.

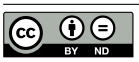
Additional Key Words and Phrases: insect swarm, collective behavior, dynamics

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1 Introduction

The dynamics of living systems, from crowds and animal herds, to insects swarms is vital for understanding and engineering our environment. Insects, and in particular ants, exhibit many similarities with humans in regards to transportation, traffic organization, collective responses in emergencies, and other factors. Furthermore, insects, and in particular ants, exhibit unique abilities to self-assemble into animate, living structures. Several such structures demonstrate unique material properties by displaying simultaneous fluid and solid-like behaviors. Due to the unique properties of such living structures, there has been much research into their mechanics, with

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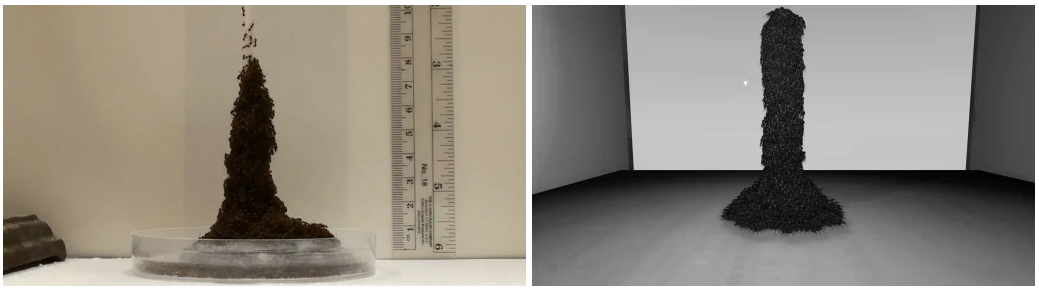


Fig. 2. Ant tower. Real world experiment on the left [Phonekeo et al. 2017], compared with our method.

potential applications in biology, engineering and materials science. In this paper, our goal is to simulate such ant aggregations.

Ant aggregations have shown multiple promising attributes which can be helpful for developing materials that can self-assemble and self-heal [Wagner et al. 2024]. For example, after cutting a high-density blob of ants with a knife, the blob will soon regain its shape. Despite much progress in our understanding of the physics behind such aggregations, simulating such interactions virtually has been largely overlooked in graphics, due to the inherent complexity in capturing both individual and collective motions simultaneously.

Approaching this domain from a physics-based, particle perspective, ants are not passive spectators, but change and move in response to their environment as active matter [Hu et al. 2016]. Due to such active matter properties, prior work which has simulated "regular" materials is not directly transferable to this target domain.

Collective, multi-agent dynamics approaches are also not adequate. Recent computer graphics work that focused on simulating insects and ants at-large does not model aggregations of crawling insects such as ants in high densities [Guo et al. 2017]. Flying insects, for example, have different dynamics since flying insects can naturally spread out in the air [Chen et al. 2019]. In another example, ant simulations have focused on goal-oriented behaviors on a 2D plane, such as food foraging, which does not typically involve high densities of ants climbing on top of each other [Xiang et al. 2019]. To the best of our knowledge, prior work in this domain has not readily simulated 3D ant aggregations, where individuals can join their bodies together, and display novel group-based material properties.

Our main contributions are: (1) we propose a novel approach for modeling 3D ant aggregations, which allows us to realistically capture such dynamics interactively for thousands of individuals. This is achieved via a novel fusion of position-based dynamics constraints (Section 3); (2) we demonstrate our method by comparing directly with real-world reference experiments and show that we capture similar physical and material qualities (Section 4).

To the best of our knowledge, our method is the first to simulate 3D ant aggregations and represents the first step toward realistic simulation of various insect aggregations based on the collective knowledge developed in biology and other scientific fields.

2 Related Work

Our work is related to physics-based simulation at-large and the simulation of collective dynamics in other related areas. We first broadly survey the above mentioned areas, before discussing our most-related work.

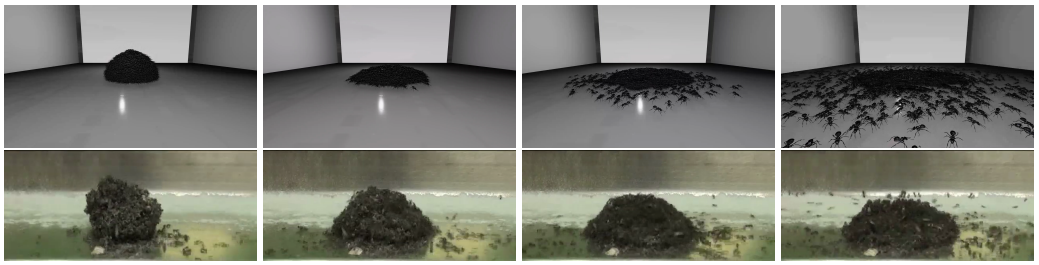


Fig. 3. Dispersal. Left to right: a blob of ants is placed in the center, after which it slowly disperses across the experimental area over time. Top: our method. Bottom: reference experiment [Tennenbaum et al. 2016].

2.1 Physics-Based Simulation

Physics-based modeling techniques have been used for simulating multiple natural phenomena, such as snow, water and elastic objects. Most related to our work are methods that simulate phenomena with a particle-focused scheme for deformable materials [Clavet et al. 2005].

In regards to ant swarms, the overall group behavior resembles a material that is both viscous and elastic [Tennenbaum et al. 2016]. Researchers have proposed various methodologies to simulate materials such as the Material Point Method [Fang et al. 2019; Ram et al. 2015] and Smoothed-Particle Hydrodynamics (SPH) [Koschier et al. 2022; Zhang et al. 2024], which is a particle-based approach closely aligned with our method. SPH is known for simulating continuum media such as fluids [Koschier et al. 2022], using forces that act on fluid particles to enforce certain properties, such as incompressibility, density, and viscosity. For example, viscosity is modeled by adding a force that damps velocity differences between neighboring particles, simulating internal friction.

Our method is based on PBD, a particle-based method [Barreiro et al. 2017; Macklin et al. 2014; Takahashi et al. 2016]. PBD was originally proposed as a means of simulating interactive physics in games, where speed and robustness takes priority over physical realism [Müller et al. 2007]. Researchers have used PBD to simulate deformable materials, fluids, and other phenomena [Bender et al. 2017]. Contrary to SPH, PBD methods work directly with positions, iteratively adjusting particles' predicted positions to satisfy a set of constraints.

2.2 Collective Dynamics

Insect swarms simulations are part of an overarching area of collective dynamics simulations, which include crowds [van Toll and Pettré 2021], robots [Van Den Berg et al. 2011], animals of various species, and other entities that exhibit group dynamics [Weiss 2023]. Early work by Reynolds and colleagues [Reynolds 1987], nicknamed the Boids model, focused on visualizing the collective behavior of fish schools and bird flocks. Boids calculates agent movements based on forces defined as functions of agent-agent distances, orientations, collisions, and navigation goals. Other researchers have built on this work by adding more accurate agent models [Ishiwaka et al. 2022; Satoi et al. 2016; Tan et al. 2011] and relying on data [Ecormier-Nocca et al. 2019; Xiang et al. 2020].

Researchers have focused on simulating various flying insects such as bees [Chen et al. 2024], flies, butterflies, and locusts [Li et al. 2015; Ren et al. 2016; Wang et al. 2015]. Forces that move individual insects, in conjunction with a flow field, are important factors for creating such simulations. Another factor is noise, since flying insects move in a noisy-looking manner. For example, curl-noise was used to simulate insects flying in a constrained shape [Chen et al. 2019].

Regarding ant simulation, most prior work has focused on simulating ant movement on a 2D plane. [Xiang et al. 2019] proposed a method for simulating ant swarms moving on a plane. The

Algorithm 1 Ant Aggregation Mechanics Simulation Loop

for each particle i **do**
  calculate $\boldsymbol{v}_i$ from a velocity planner ▷ §3.2
  $\boldsymbol{x}_i^* \leftarrow \boldsymbol{x}_i + \Delta t \boldsymbol{v}_i$
end for
Compute nearest neighbors ▷ §3.1
while iter < SolverIterations **do**
  Apply density constraints **for each** particle ▷ §3.3
  Apply viscosity **for each** particle ▷ §3.4
  Project distance constraints ▷ §3.5, 3.6
  Resolve particle-particle collisions with friction ▷ §3.7
  Keep in bounds **for each** particle ▷ §3.8
end while
Resolve particle-rigid body collisions ▷ §3.8
for all particles i **do**
  $\boldsymbol{v}_i \leftarrow \text{clamp}(s_{max}, k_{\text{limit}}(\boldsymbol{x}_i^* - \boldsymbol{x}_i)/\Delta t)$ ▷ §3.9
  $\boldsymbol{x}_i \leftarrow \boldsymbol{x}_i^*$
end for

study focused on a birds-eye view visualization, without inter-ant contact mechanics and 3D movement. [Guo et al. 2017] simulated collective transport of ants moving objects. Others proposed capturing ant swarm behaviors via robots. For example, Mayet and colleagues focused on emulating ant pheromone trail behavior [Mayet et al. 2010], and Rubenstien et al. proposed to use robots for mimicking ant self-assembly behaviors [Rubenstein et al. 2014]. Despite much progress in simulating ant swarms, there has not been work on accurately simulating ant swarm mechanics in 3D aggregations to the best of our knowledge.

3 Method

Our approach for simulating ant aggregations is inspired by Position-Based Dynamics [Müller et al. 2007]. A traditional PBD solver steps from time n to time $n+1$ with a forward Euler position prediction: $\boldsymbol{x}_i^* = \boldsymbol{x}_i^n + \Delta t(\boldsymbol{v}_i^n + \Delta t \boldsymbol{f}_{\text{ext}}(\boldsymbol{x}_i^n))$, where $\boldsymbol{f}_{\text{ext}}$ represents external forces such as gravity. Without any further interference, particles would advect in the velocity field, ignoring other particles. To amend such nonphysical behavior, PBD defines constraint functions, which control the dynamics of the simulation by augmenting particle positions.

In our method, individual ants are modeled as spherical particles, similar to PBD definitions [Müller et al. 2007]. Ant dynamics are influenced by constraints, which are associated with:

- a scalar function $C \in \mathbb{R}$,
- a stiffness parameter $k \in [0, 1]$, and
- a constraint type,

which is either an *equality* $C(\boldsymbol{x}^*) = 0$ or an *inequality* constraint, where $C(\boldsymbol{x}^*) \geq 0$, and $\boldsymbol{x}^* = [\boldsymbol{x}_1^{*T}, ..., \boldsymbol{x}_n^{*T}]^T$ is a vector of predicted positions of the particles involved in the constraint.

Each ant has the following properties:

(1) a position $\boldsymbol{x}_i \in \mathbb{R}^3$,
(2) a predicted position $\boldsymbol{x}_i^* \in \mathbb{R}^3$,
(3) a velocity $\boldsymbol{v}_i \in \mathbb{R}^3$,
(4) a radius $r_i \in \mathbb{R}^+$, and

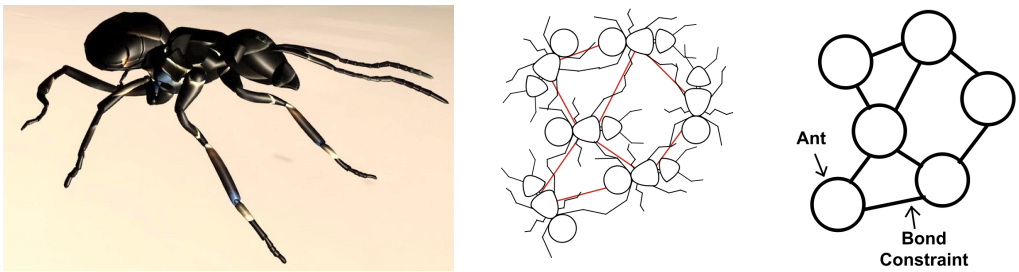


Fig. 4. Illustration of ant bonds. Left: 3D render of an ant. Middle: Illustration of bond connections between neighboring ants. Right: Representation of the connections as distance constraints between particles.

(5) a mass m_i, and corresponding inverse mass $w_i = 1/m_i \in \mathbb{R}^+$.

PBD utilizes positional corrections $\Delta \boldsymbol{x}$ which satisfies the above mentioned constraints. e.g. $C(\boldsymbol{x}^* + \Delta x) = 0$. This equation can be approximated by:

$$C(\boldsymbol{x}^* + \Delta \boldsymbol{x}) \approx C(\boldsymbol{x}) + \nabla_{\boldsymbol{x}^*} C(\boldsymbol{x}^*) \Delta \boldsymbol{x}, \quad \text{with } \Delta \boldsymbol{x} = \lambda \nabla_{\boldsymbol{x}^*} C(\boldsymbol{x}^*), \tag{1}$$

where Δx is the positional correction restricted along the direction of the constraint gradient, and λ is a scaler coefficient defined with:

$$\lambda = -\frac{C(\boldsymbol{x}_1^*, \ldots, \boldsymbol{x}_n^*)}{\sum_j ||\nabla_{\boldsymbol{x}_j^*} C||^2}. \tag{2}$$

Below we summarize our adaptation of the PBD method and detail our solver and constraints.

3.1 Solver

Constraints update the particle's predicted position, in a Gauss-Sidel fashion [Müller et al. 2007]. Unless noted otherwise, all particles participate in all constraints. If no stiffness parameter is associated with a constraint, the stiffness is set to 1. We randomize the initial rotation for each ant in regards to the closest surface plane, and all particles are assigned a mass $m = 1$. In the start of the timestep, we apply forces with F_{ext}. This is usually gravity, except in cases in which ants self-actuate to explore an area (Fig. 3,6). When particles touch the floor, we add friction to reduce the speed of their movement.

For detecting nearest neighbors, we use a spatial hash with uniform-sized cells [Clavet et al. 2005]. Each cell is a cube which stores a list of particles. When searching for neighbors for a specific particle, we search in a 3D window of cubes around the cube that is associated with such particle, until a predefined distance threshold $d_{\text{neighbors}}$.

3.2 Velocity Planner

Each ant is influenced by gravity g and by random exploration movement [Popp and Dornhaus 2023]:

$$v_i \leftarrow v_i + \Delta t g + v_{\text{random}}, \quad v_{\text{random}} = [\cos\theta, \sin\theta] s_i,$$

where $\theta \in [-10, 10]$ is a uniformly sampled random angle in degrees, $g = 280$ is the gravity force, $s_i = 120$ is a predefined preferred speed, and v_{random} is parallel to the ant's current ground surface. Exploration can be turned on or off for each experiment. In case it is on, exploration occurs when for ant i when it is on a surface, and the ant is not inhibited by other ants, i.e. the ant is not inside an aggregation nor bond-constrained to an existing ant or surface. In case the ant is not exploring, we set $s_i = 0$.

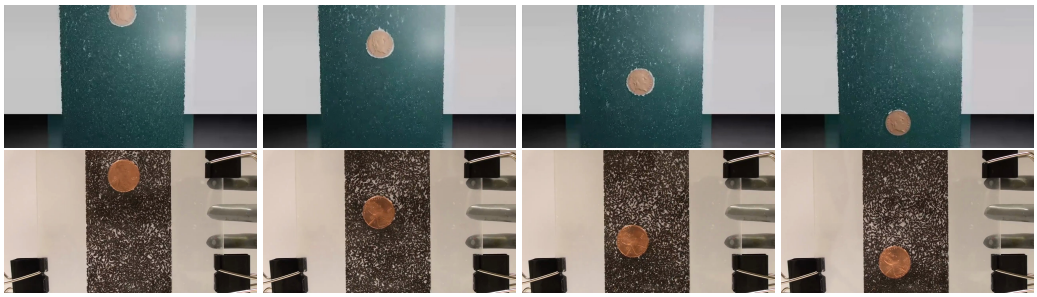


Fig. 5. Coin drop. A coin is dropped through a rectangular vessel filled with ants, which is about 2 ants thick. Our method (top) closely mimics a real-world reference experiment (bottom).

3.3 Density

At high densities, aggregations of several ant species behave as a viscoelastic material, exhibiting both solid and fluid-like properties [Hu et al. 2016; Tennenbaum et al. 2016]. We emulate fluid-like responses by maintaining a consistent density across the aggregation, which we measure with the density estimator:

$$\rho_i = \sum_{j \in N} m_j W(\boldsymbol{x}_i^* - \boldsymbol{x}_j^*, h), \tag{3}$$

where j iterates over all neighboring particles, W stands for the density estimation kernel, and h being the width of the kernel. We use the Poly6 kernel for density estimation and the Spiky kernel for gradient calculation [Müller et al. 2003].

In 2013, Macklin and colleagues proposed a constraint for maintaining a user-defined density ρ_0 [Macklin and Müller 2013]:

$$C_i(\boldsymbol{x}_1^*, \ldots, \boldsymbol{x}_n^*) = \frac{\rho_i}{\rho_0} - 1 = 0, \text{ s.t. } \nabla_{\boldsymbol{x}_i} C = \frac{1}{\rho_0} \sum_j m_j \nabla_{\boldsymbol{x}_i} W(\boldsymbol{x}_i^* - \boldsymbol{x}_j^*, h), \tag{4}$$

where $1, \ldots, n$ are the particles participating in the constraint, ρ_0 is the rest density and ρ_i is calculated using the standard SPH density estimator (Eq. 3). We activate this constraint for ant i only when the ant is not exploring (Section 3.2), and when there are at least $|N| \geq 15$ neighbors.

The positional corrections and scaling factor is:

$$\Delta \boldsymbol{x}_i = \frac{1}{\rho_0} \sum_j (\lambda_i + \lambda_j) \nabla W(\boldsymbol{x}_i^* - \boldsymbol{x}_j^*, h), \quad \lambda = -\frac{C(\boldsymbol{x}_1^*, \ldots, \boldsymbol{x}_n^*)}{\sum_j ||\nabla_{\boldsymbol{x}_j^*} C(\boldsymbol{x}_1^*, \ldots, \boldsymbol{x}_n^*)||^2 + \epsilon},$$

where j iterates over all neighboring particles N, λ_i is the scaling correction for the particle i, and λ_j is the scaling correction for neighboring particles j, and ϵ is a user defined relaxation parameter that is constant over the simulation, as suggested in Position-based Fluids [Macklin and Müller 2013]. Note that such definitions do not follow default PBD conventions[1] (Eq. 1).

3.4 Viscosity

The prior density constraint injects particles with a fluid-like movement quality. However, ant aggregations exhibit a viscous quality [Tennenbaum et al. 2016], where clumps of ants move

[1] A straightforward derivation should yield the correction $\Delta x_i = \frac{\lambda}{\rho_0} \sum_j m_j \nabla W(\boldsymbol{x}_i^* - \boldsymbol{x}_j^*, h)$, where λ includes all particles participating in the constraint, multiplied by the missing mass m_j.

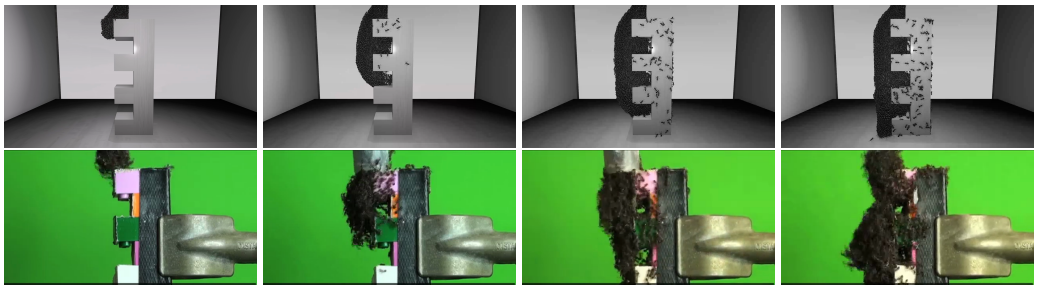


Fig. 6. Bracket. Left to right: A flow of ants coming from above behaves in a viscous manner. Groups of ants in clusters slide into cavities between bracket teeth, while several ants explore the external surfaces. Top: our method. Bottom: reference experiment [Tennenbaum et al. 2016].

together in a coherent motion. Viscosity has the affect of slowing down a material's motion. To apply viscosity, we use the following positional corrections [Müller et al. 2003]:

$$\Delta \boldsymbol{x}_i = \mu_{\text{visc}} \sum_j \frac{\boldsymbol{v}_j - \boldsymbol{v}_i}{\rho_j} \nabla^2 W(\boldsymbol{x}_i^* - \boldsymbol{x}_j^*, h), \quad \nabla^2 W(r,h) = \frac{45}{\pi h^6}(h-r), \tag{5}$$

where $\boldsymbol{v}$ are the particles velocities set in the beginning of the timestep (Alg. 1), μ_{visc} is the viscosity strength, and:

$$W(r,h) = \frac{15}{2\pi h^3}\left(-\frac{r^3}{2h^3} + \frac{r^2}{h^2} + \frac{h}{2r} - 1\right), \quad 0 \le r \le h,$$

is the viscosity kernel given in [Müller et al. 2003].

This correction modifies each ant's predicted position to comply with the overall movement direction of its neighbors, i.e. looking at the neighbors of particle i from i's own moving frame of reference, particle i is accelerated in the direction of the relative speed of its neighbors. Similar to the density constraint (Section 3.3), we activate this constraint for ant i only when the ant is not exploring, and when there are at least $|N| \ge 15$ neighboring particles.

3.5 Ant Bonds

In high-density aggregations, ants form bonds with neighboring ants via their limbs [Tennenbaum et al. 2016; Vernerey et al. 2018]. Compression of such limbs results in an elastic-like bond (Figure 4). We use PBD to model such elasticity with distance constraints of the form [Müller et al. 2007]:

$$C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) = \|\boldsymbol{x}_{ij}^*\| - d_{\text{bond}} = 0, \tag{6}$$

where $\boldsymbol{x}_{ij} = \boldsymbol{x}_i - \boldsymbol{x}_j$, and d_{bond} represents the desired rest distance between ants i and j. PBD resolves such distance constraints with a positional correction:

$$\Delta \boldsymbol{x}_i = -k_c \frac{w_i}{w_i + w_j} C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) \frac{\boldsymbol{x}_{ij}^*}{||\boldsymbol{x}_{ij}^*||}, \quad \Delta \boldsymbol{x}_j = k_c \frac{w_j}{w_i + w_j} C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) \frac{\boldsymbol{x}_{ij}^*}{||\boldsymbol{x}_{ij}^*||}, \tag{7}$$

where k_c denotes the stiffness (Table 2).

We **adaptively generate** connections depending on the positional relations of the ants. Assuming d_{ij} is the current distance between ants i and j, we generate a new connection between two ant particles if all of the following holds: (i) $\|\boldsymbol{x}_{ij}^*\| < d_{\text{bond}}$, (ii) there is no current bond between ants i and j, and (iii) the number of bonds both ants participate in is smaller than maximum allowed number of bonds. In the above conditions are positive, we increment the number of active bonds for both ants.

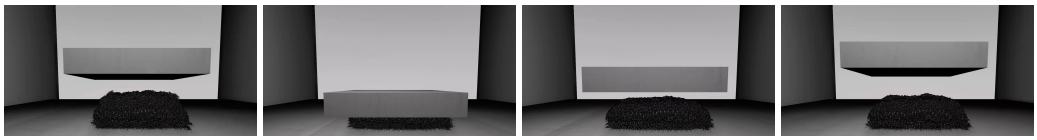


Fig. 7. A blob of ants is crushed as a plate bears down on it. Once the plate rises, the blob slowly rises to approximately match its prior shape.

We **remove** connections when one of the following conditions occur:

(1) the current distance between the ants $\|\boldsymbol{x}_{ij}^*\| > d_{\text{bond}}^{\max}$, where $d_{\text{bond}}^{\max} > d_{\text{bond}}$ is a user-defined parameter.
(2) the bond has been active for a random time t_b which is sampled from a uniform range (Table 2). This random time is re-sampled each time a bond is formed.

3.6 Adhesion

Ants have legs with a sticky quality, which allows them to easily climb and latch unto surfaces. Technically, this is referred to as an adhesive foot pad, which sticks when pulled toward the body, but unsticks when moved away from it, allowing for effortless movement [Federle and Endlein 2004]. To model such adhesion, we define the constraint:

$$C(\boldsymbol{x}_i^*, \boldsymbol{x}_{\text{surface}}) = ||x\boldsymbol{x}_i^* - \boldsymbol{x}_{\text{surface}}|| - d_{\text{adhesive}} = 0, \quad \Delta\boldsymbol{x}_i = -k_{ad} C(\boldsymbol{x}_i^*, \boldsymbol{x}_{\text{surface}}) \frac{\boldsymbol{x}_i^* - \boldsymbol{x}_{\text{surface}}}{||\boldsymbol{x}_i^* - \boldsymbol{x}_{\text{surface}}||} \tag{8}$$

where d_{adhesive} is the rest distance for the constraint, and $\boldsymbol{x}_{\text{surface}}$ is the closest point on the adhesive surface. Unlike Eq. 7, the correction depends fully on the ant – we define the surface contact point x_{surface} to have infinite mass.

We **activate** this constraint when an ant is not exploring (Section 3.2), and when an ant is touching such adhesive surface $||\boldsymbol{x}_i^* - \boldsymbol{x}_{\text{surface}}|| < d_{\text{adhesive}}$ (Table 2). We **deactivate** adhesion connections in a manner similar to the ant-ant bond constraint (Section 3.5), when either i) $||x\boldsymbol{x}_i^* - \boldsymbol{x}_{\text{surface}}|| > d_{\text{adhesive}}^{\max}$, where $d_{\text{adhesive}}^{\max} > d_{\text{adhesive}}$ is a user defined parameter, or ii) t_b seconds have passed since the initialization of the constraint, where t_b is uniform random time resampled each time an adhesion constraint is formed (Table 2).

3.7 Ant-Ant Contact

During a simulation time-step, ant particles may inadvertently overlap with each other. To detect and resolve such situations, we use PBD distance constraints:

$$C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) = \|\boldsymbol{x}_{ij}^*\| - (r_i + r_j) \geq 0, \tag{9}$$

where $\boldsymbol{x}_{ij} = \boldsymbol{x}_i - \boldsymbol{x}_j$, and r_i, r_j represent the respective ant radii. In case $C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) < 0$, a collision occurs. PBD resolves such collisions with positional corrections:

$$\Delta\boldsymbol{x}_i = -\frac{w_i}{w_i + w_j} C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) \frac{\boldsymbol{x}_{ij}^*}{||\boldsymbol{x}_{ij}^*||}, \quad \Delta\boldsymbol{x}_j = \frac{w_j}{w_i + w_j} C(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*) \frac{\boldsymbol{x}_{ij}^*}{||\boldsymbol{x}_{ij}^*||}, \tag{10}$$

We then immediately add the correction to the ants's predicted position with $\boldsymbol{x}_i^*$ and similarly for ant $\boldsymbol{x}_j^*$.

In case of a collision, we first resolve interpenetration. We then apply friction to the each ant's motion by reducing the relative tangential displacement of the particles during this time-step. Using

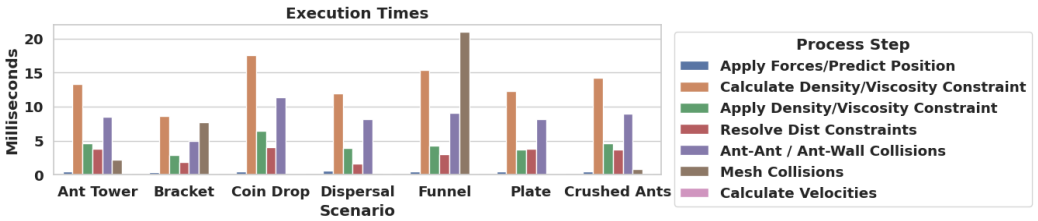


Fig. 8. Timings, not including rendering.

prior work definitions [Macklin et al. 2014]:

$$\Delta \boldsymbol{x}_{\perp} = [(\boldsymbol{x}_i^* - \boldsymbol{x}_i) - (\boldsymbol{x}_j^* - \boldsymbol{x}_j)] \perp \mathbf{n},$$

where $\mathbf{n}$ is the contact normal $n = \boldsymbol{x}_{ij}^* / ||\boldsymbol{x}_{ij}^*||$. The frictional position delta for particles i and j are then:

$$\Delta \boldsymbol{x}_i = \frac{w_i}{w_i + w_j} \Delta \boldsymbol{x}_{\perp} \mu_a, \quad \Delta \boldsymbol{x}_j = -\frac{w_j}{w_i + w_j} \Delta \boldsymbol{x}_{\perp} \mu_a,$$

where μ_a is the kinetic inter-ant friction coefficient. Note that unlike [Macklin et al. 2014], we do not model static friction due to each ant's ability to self-actuate.

3.8 Surface Contact

To prevent ants from locomoting outside the experiment area, we verify that each ant is within the experiment's bounding box. If that is not the case, we adjust the $\boldsymbol{x}_i^*$ to be the closest point within the experiment, s.t. the ant does not overlap with any wall:

$$C(\boldsymbol{x}_i^*, \boldsymbol{x}_{\text{surface}}) = ||\boldsymbol{x}_i^* - \boldsymbol{x}_{\text{surface}}|| - r_i > 0, \tag{11}$$

where $\boldsymbol{x}_{\text{surface}}$ is the closest boundary point, and r_i is the ant's representative particle radii.

For ant collisions with rigid bodies, we use a constraint similar to the one proposed in [Müller et al. 2007], similar to Eq. 11. For each ant, we use a test ray starting from the particle's position $\boldsymbol{x}_i^*$ to check whether the ant is inside an object. If that is the case, we find the closest corresponding triangle on the object by iterating over all such elements. Given the closest triangle, we compute a surface point outside the object, which is normal to the triangle, and set $\boldsymbol{x}_i^*$ to such point.

3.9 Friction and Velocity Limiting

After the constraint solve, we limit the speed of ants to better approximate real world movement capabilities. We first apply $\boldsymbol{v}_i = (\boldsymbol{x}_i^* - \boldsymbol{x}_i)(0.999 - s_{\text{damping}})/\Delta t$, where $s_{\text{damping}} \in [0, 1]$ can be

Table 1. Experiments, where k_{ad} referes to surface adhesive constraint strength (Sec. 3.6), and s_{surface} refers to surface friction (Sec. 3.9), and exploration denotes whether exploration is on or off for the ant movement (Sec. 3.2). We used 8,500 particles for all experiments.

Scenario	k_{ad}	s_{surface}	Exploration?	**FPS**
Funnel (Fig. 9)	-	0.024	yes	18.9
Bracket (Fig. 6)	0.00095	0.208	yes	51.062
Plate (Fig. 10)	0.1072	0.998	no	33.319
Disperse (Fig. 3)	-	0.208	yes	36.079
Coin Drop (Fig. 5)	-	0.208	no	24.227
Ant Tower (Fig. 2)	0.0172	0.208	no	28.835
Crush (Fig. 5)	-	0.998	no	28.887

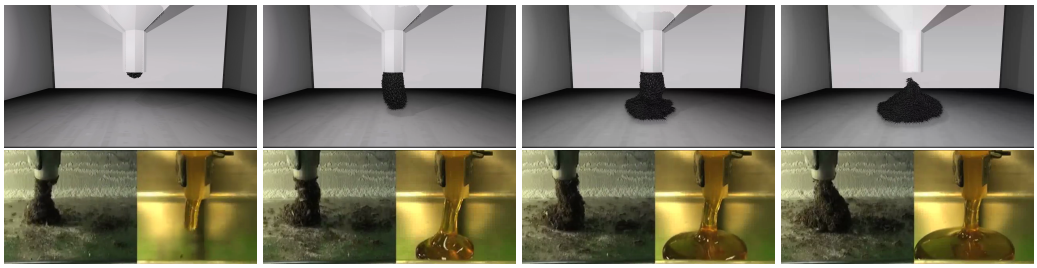


Fig. 9. Funnel. An aggregation of ants is placed on top of a funnel, and then compresses to flow downward through a narrow opening. Top: our method, bottom: reference experiment [Tennenbaum et al. 2016], where ant flow is compared with honey.

viewed as either a damping or drag coefficient to model both inter-ant, and environmental friction. The value s_{damping} depends on the surface the ant has been in contact with during a time-step. In our experiments this is either the floor s_{floor} or the experimental prop in the middle of the scene s_{surface}. If an ant has not contacted any surface, e.g. it is inside an aggregation, we set $s_{\text{damping}} = 0$. Additionally, we clamp the speed of ants to be $||\boldsymbol{v}_i|| < V_{\max}$, to emulate biomechanical limitations on the motion of ants. Note that at the end of the iteration, each ant's position is updated to its predicted position, regardless of the velocity modification above. The intent here is to preserve the full positional correction, whilst preserving biomechanically realistic speeds, similar to PBD crowds [Weiss et al. 2019].

4 Results

We evaluate the performance of our method on a variety of aggregation scenarios to test the strength of our approach versus reference experiments with live ants. We use Unity for rendering, where each particle is represented as an animated ant with a simple walking animation. We also globally scaled the location of the ants for better visualization of the aggregations.

We implemented our method in C++, where specific components are parallelized with OpenMP – including nearest neighbors search (Section 3.1), the fluid and viscosity constraints (Section 3.3,3.4), and mesh collisions (Section 3.8). In terms of performance, we collected metrics to evaluate the computational cost of our method and its various components, see Figure 8. All experiments were run on a Intel i7-11370H machine. Our simulations include thousands of particles, and as shown in Figure 8, our method is interactive in all scenarios, with the largest computation cost being the density and viscosity constraints, followed by collision. We also tested the affect of various constraints being active or not, see the supplemental video for more information. Note that our implementation is not as scalable as a GPU one [Macklin et al. 2014]. Regardless, further code optimization is possible for decreasing computational costs. Our implementation is readily available in an open source repository[2].

4.1 Ant Tower

Ants were recorded to build towers and we used our method to capture such abilities (Figure 2). The ants surround and hang off a pillar using adhesive bonds. Adhesive strength is strong enough to prevent the ants from sliding down the pillar, but flexible enough to allow some slight movement (Section 3.6). The ants are spawned in a rectangular prism shape around the pillar and a spherical shape near the floor.

[2](**https://github.com/mwl-6/AntSwarmDynamics**)

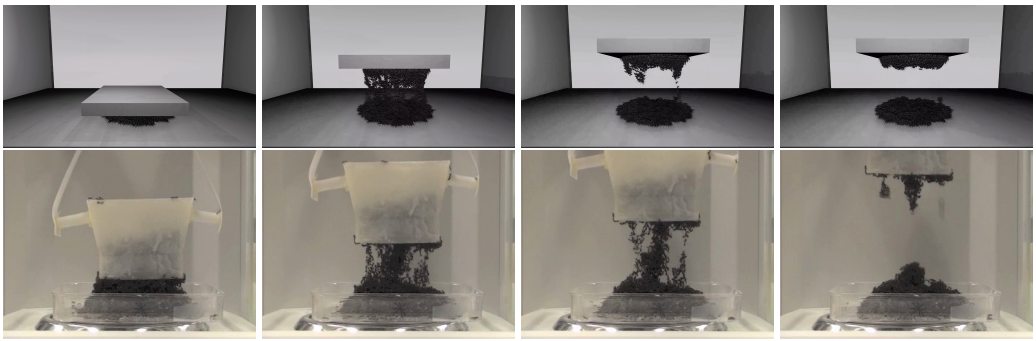


Fig. 10. Plate contracting an ant blob. Left to right: when the plate lifts, ants holds their limbs together, which results in an elastic-like material. Eventually, the links rupture due to excessive stretching. Top: our method. Bottom: reference experiment [Phonekeo et al. 2016].

4.2 Bracket

The bracket scene begins with ants gradually pouring onto the top of a bracket (Figure 6). Over time, the ants begin to pile up and fall on the side of the bracket. As the ants approach a gap, they begin to move inside and fill the space. The main body of ants continues down the side of the bracket. Ants pushed against the surface of the bracket are slowed by friction and attach themselves using adhesion bonds (Section 3.6). The rest of the body of ants, linked together using distance constraints, are then affected as a result (Section 3.5). Their descent is slowed as well. Over time, some ants break their connection to the main body of ants and wander off on their own. They crawl around the surface of the bracket. The main body of ants continues down the bracket at a slow speed, filling gaps along the way until the ants reach the bottom. In this scenario, we set the preferred speed $s = 10$ when ants traverse the bracket, and use the default value when for each ant that reaches the floor. We also disable ant-wall bonds from disconnecting.

4.3 Coin Drop

In this experiment, we show how ants behave as a viscous fluid by simulating a coin falling through a vessel containing ants. Note that the coin does not fall rapidly due to the ant aggregations locked into the vessel (Figure 5). As the coin sinks, it pushes the ants around it. Ant-coin collisions are resolved in a way that is similar to ant-ant collisions, only instead of equal positional corrections for both colliding parties, the coin is assigned a mass roughly 4500 times larger than that of an ant (Eq. 10), which forces ants to makes most of the positional corrections.

4.4 Dispersal

This setting demonstrates how ants will disperse around the experimental medium, starting from a blob placed in the center of the experimental area (Figure 3). The scene begins with a pile of ants held together by distance constraints. Over time, their bonds dissolve and the ants are free to move (Section 3.5). As the ants move independently, the pile spreads out and disappears. The ants move in a random direction given an angle, which can rotate left or right randomly over time.

4.5 Funnel

In this experiment, we demonstrate the viscous behavior of an ant blob as it flows down a funnel (Figure 9). A pile of ants slowly squeezes through a narrow funnel in a tightly packed manner, before forming a pile on the ground. Viscosity helps recreate the fluid-like behavior of the ants

(Section 3.4). Once reaching the ground, the ants break their bonds over time and explore randomly similar to the dispersal experiment.

4.6 Plate

This experiment demonstrates the elastic effects formed by bonding ants (Figure 10). A plate descends and crushes a pile of ants on the floor. The ants on top of the pile stick to the surface of the plate. As the plate rises, ants are pulled with it. The rest of the pile, connected through distance constraint bonds, are pulled upwards too (Section 3.5). The pile is stretched by the ants adhering to the plate and the ants adhering to the floor. Eventually the bonds snap and the pile is split in two. The scene ends with some ants hanging from the bottom of the plate. Adhesive strength is stronger, forcing the ants to adhere to either the floor or the plate. As a result, the pile will eventually split once it is stretched to its breaking point.

4.7 Crush

The second plate experiment is similar to the first, but omits the adhesive strength setting. A blob of ants is crushed by a descending plate, which shortly thereafter rises. As the plate rises, the blob expands to regain lost volume (Fig. 7). Here we used a combination of density and distance constraints which allows for an object to be deformed but return its original shape.

5 Discussion

In this work, we proposed a method for simulating the mechanics of ant aggregations. We demonstrated our method in several examples that mirror real-world experiments. In all settings, our method runs interactively, and can also be easily integrated with recent game and simulation frameworks that use PBD. To the best of our knowledge, our method is the first to simulate such 3D aggregations virtually.

Parameter	Value	Refer to
Δt Timestep [sec]	0.01	Alg. 1
Solver iterations	4	Alg. 1
r radius	7	Sec. 3
ρ_0 Ideal density	1e−5	Sec. 3.3
ϵ Density relaxation	1	Sec. 3.3
h Neighbor search distance	60	Sec. 3.3
μ_{visc} Viscosity	0.408	Sec. 3.4
Max # bonds	3	Sec. 3.5
d_{bond} Rest distance for ant-ant bonds	$4.28r$	Sec. 3.5
$d_{\text{bond}}^{\text{max}}$ Bond threshold distance	$2.84d_{\text{bond}}$	Sec. 3.5
k_c Bond stiffness	0.0308	Sec. 3.5
t_b Time to server bonds [sec]	$t_b \sim \mathcal{U}(1.8, 3.65)$	Sec. 3.5
d_{adhesive} Rest distance for adhesion	$0.428r$	Sec. 3.6
$d_{\text{adhesive}}^{\text{max}}$ Adhesive threshold distance	$14.2d_{\text{adhesive}}$	Sec. 3.6
μ_a Ant-ant contact friction	0.1512	Sec. 3.7
s_{floor} Floor friction	0.8	Sec. 3.9

Table 2. Default parameters. All experiments use the above listed values, unless noted otherwise.

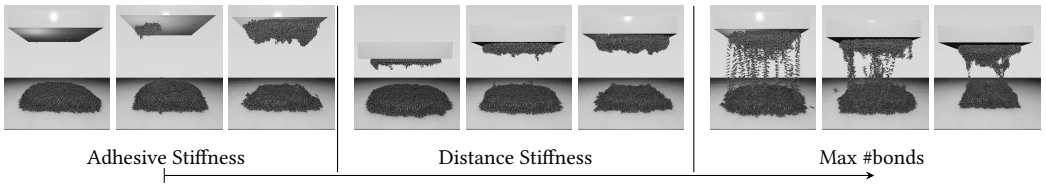


Fig. 11. Effect of increasing parameter values from left to right in the Plate experiment (Sec. 4.6). Stronger adhesion results in additional ants attaching to the rising plate. Larger ant-ant bond stiffness results in tighter and larger ant concentrations that are capable of self-preservation after being stretched or crushed. Increasing the maximum number of bonds per ant can also achieve a similar effect. Note that a low number of bonds results in a stringy-like look for the aggregation.

5.1 Parameter Analysis

Our method uses a set of parameters that can be adjusted to portray different aggregation characteristics. We discuss the effect of changing several parameter values:

- **Friction**: frictional grip is crucial for ants to climb and move on surfaces. Without this parameter, ant particles will slide on surfaces (see supplemental video). Increasing friction values inhibits movement momentum.
- **Ant-ant bonds**: are determined by the bond rest distance, stiffness, and number of allowed bonds — all which influence the appearance and density of an aggregation. For example, increasing the rest distance results in a more spread out pile. Higher stiffness values and allowed number of bonds results in ants forming tighter clusters (Fig. 11).
- **Adhesive stiffness**: stronger adhesion allows particles to cling to nearby surfaces, resist other forces, and restrain the movement of the rest of the aggregation. A weaker adhesive strength may allow for particles to resist movement away from a surface, but it will also be easier to break surface bonds using other forces such as ant bonds (Fig. 11).
- **Kernel size**: a reduced kernel size can lead to jittery surfaces. It can also lead to erratic behavior where particles shoot away from other particles at high velocities. Increasing kernel size does not change visuals or affect stability much, but greatly slows down performance.

5.2 Method Scope and Limitations

Our work has several limitations:

- *Animation*: We focused on capturing the mechanics of ant aggregations. Enhancing the movements or appearance of the ants is important, but orthogonal to our contribution.
- *Realism*: We achieve visual realism between the results produced by our method and real-world footage. Metrics produced by real-world experiments vary, and pertain to specific ants species, such as fire ants, weaver ants, and similar, where each species may exhibit different aggregation properties. Here our focus is a general simulation framework for generic ant aggregations.
- *Mechanical properties*: In certain densities and configurations, several ant species aggregations have been shown to behave similar to shear thinning fluids [Tennenbaum et al. 2016]. We experimentally found that PBD constraints [Macklin and Müller 2013] that are attuned to Newtonian fluids are sufficient for visual realism between the real-world and virtual experiments.
- *PBD and XPBD*: XPBD allows more fine grain control over constraint stiffness, disentangling it from the solver's number of iterations [Macklin et al. 2016]. In our experiments, we found default PBD [Müller et al. 2007] sufficient for simulating various ant aggregations scenarios.
- *Targeted Locomotion*: Many prior works focused on ant swarm behaviors on a 2D plane, ranging from food foraging [Xiang et al. 2019] to collective transport [Guo et al. 2017]. Such behaviors

require intelligently controlling each ant's velocity on a 2D plane for specific, pre-determined locomotion goals. Capturing such targeted locomotion aspects is not the focus of our work — we do not encode any high-level behavioral motion patterns. Our focus is on the general 3D mechanics of ant aggregations.

6 Future Work

There are several avenues for future work. First, our current implementation is partially parallelized via OpenMP and runs on the CPU. In terms of computational speed, our implementation is slower than previous PBD work that focused on simulating physical phenomena without insect aggregation behaviors [Macklin et al. 2014]. For full parallelization, we will most likely need to adopt a Jacobi solver and use a GPU language such as CUDA, which requires parallel computing support via Kernel-style computations [Macklin et al. 2014]. Such machinery will allow supporting at least tens of thousands of entities interactively. Second, we have only simulated a subset of all potential ant aggregation phenomena. We are interested in integrating higher-level swarm behaviors, allowing for the simulation of different collective dynamics and objectives. Such behaviors include path-finding using pheromones, trail following, nest construction, collective search, bridge self-assembly, implicit communication for collision-free movement, thigmotaxis, and self-regulation of density for traffic flow. We leave such phenomena for future work [Graham et al. 2017; Phonekeo et al. 2017; Reid et al. 2015]. Third, we currently model ants as spherical particles. While spheres serve only as an approximate estimator of each ant's position, it may be more accurate to approximate ants with multiple spheres or even with elliptical particles [Müller and Chentanez 2011; Talukdar and Weiss 2025]. Finally, in nature, there are many other insects and entities which form aggregations [Hu et al. 2016]. We believe that realistic visual simulations of such phenomena may increase our understanding and open new opportunities in the intersection of engineering and biology. We also hope that industry practitioners can build on our method to portray more physically accurate swarms visual effects, which have seen usage in recent films [Devlin et al. 2023; Pražák et al. 2018].

References

Héctor Barreiro, Ignacio García-Fernández, Iván Alduán, and Miguel A Otaduy. 2017. Conformation constraints for efficient viscoelastic fluid simulation. *ACM Transactions on Graphics (TOG)* 36, 6 (2017), 1–11.

Jan Bender, Matthias Müller, and Miles Macklin. 2017. A survey on position based dynamics, 2017. *Proceedings of the European Association for Computer Graphics: Tutorials* (2017), 1–31.

Qiang Chen, Wenxiu Guo, Yuming Fang, Yang Tong, Tingsong Lu, Xiaogang Jin, and Zhigang Deng. 2024. A Bio-inspired Model for Bee Simulations. *IEEE Transactions on Visualization and Computer Graphics* (2024).

Qiang Chen, Guoliang Luo, Yang Tong, Xiaogang Jin, and Zhigang Deng. 2019. Shape-constrained flying insects animation. *Computer Animation and Virtual Worlds* 30, 3-4 (2019), e1902.

Simon Clavet, Philippe Beaudoin, and Pierre Poulin. 2005. Particle-based viscoelastic fluid simulation. In *Proceedings of the 2005 ACM SIGGRAPH/Eurographics symposium on Computer animation*. 219–228.

Nathan Devlin, Yasser Hamed, Alberto Luceño Ros, Jeff Sullivan, and D'Lun Wong. 2023. Creating Creature Chaos: The Methods that Brought Crowds to the Forefront on Disney's. In *ACM SIGGRAPH 2023 Talks*. 1–2.

Pierre Ecormier-Nocca, Julien Pettré, Pooran Memari, and Marie-Paule Cani. 2019. Image-based authoring of herd animations. *Computer Animation and Virtual Worlds* 30, 3-4 (2019), e1903.

Yu Fang, Minchen Li, Ming Gao, and Chenfanfu Jiang. 2019. Silly rubber: an implicit material point method for simulating non-equilibrated viscoelastic and elastoplastic solids. *ACM Transactions on Graphics (TOG)* 38, 4 (2019), 1–13.

Walter Federle and Thomas Endlein. 2004. Locomotion and adhesion: dynamic control of adhesive surface contact in ants. *Arthropod Structure & Development* 33, 1 (2004), 67–75. https://doi.org/10.1016/j.asd.2003.11.001 Attachment Systems of Arthropods.

Jason M Graham, Albert B Kao, Dylana A Wilhelm, and Simon Garnier. 2017. Optimal construction of army ant living bridges. *Journal of theoretical biology* 435 (2017), 184–198.

Shihui Guo, Meili Wang, Gabriel Notman, Jian Chang, Jianjun Zhang, and Minghong Liao. 2017. Simulating collective transport of virtual ants. *Computer Animation and Virtual Worlds* 28, 3-4 (2017), e1779.

DL Hu, S Phonekeo, E Altshuler, and F Brochard-Wyart. 2016. Entangled active matter: From cells to ants. *The European Physical Journal Special Topics* 225 (2016), 629–649.

Yuko Ishiwaka, Xiao S Zeng, Shun Ogawa, Donovan Westwater, Tadayuki Tone, and Masaki Nakada. 2022. DeepFoids: Adaptive Bio-Inspired Fish Simulation with Deep Reinforcement Learning.. In *NeurIPS*.

Dan Koschier, Jan Bender, Barbara Solenthaler, and Matthias Teschner. 2022. A survey on SPH methods in computer graphics. In *Computer graphics forum*, Vol. 41. Wiley Online Library, 737–760.

Weizi Li, David Wolinski, Julien Pettré, and Ming C. Lin. 2015. Biologically-inspired visual simulation of insect swarms. In *Computer Graphics Forum*, Vol. 34. Wiley Online Library, 425–434.

Miles Macklin and Matthias Müller. 2013. Position based fluids. *ACM Transactions on Graphics (TOG)* 32, 4 (2013), 1–12.

Miles Macklin, Matthias Müller, and Nuttapong Chentanez. 2016. XPBD: position-based simulation of compliant constrained dynamics. In *Proceedings of the 9th International Conference on Motion in Games*. 49–54.

Miles Macklin, Matthias Müller, Nuttapong Chentanez, and Tae-Yong Kim. 2014. Unified particle physics for real-time applications. *ACM Transactions on Graphics (TOG)* 33, 4 (2014), 1–12.

Ralf Mayet, Jonathan Roberz, Thomas Schmickl, and Karl Crailsheim. 2010. Antbots: A feasible visual emulation of pheromone trails for swarm robots. In *International conference on swarm intelligence*. Springer, 84–94.

Matthias Müller, David Charypar, and Markus Gross. 2003. Particle-based fluid simulation for interactive applications. In *Proceedings of the 2003 ACM SIGGRAPH/Eurographics symposium on Computer animation*. Citeseer, 154–159.

Matthias Müller and Nuttapong Chentanez. 2011. Solid simulation with oriented particles. In *ACM SIGGRAPH 2011 papers*. 1–10.

Matthias Müller, Bruno Heidelberger, Marcus Hennix, and John Ratcliff. 2007. Position based dynamics. *Journal of Visual Communication and Image Representation* 18, 2 (2007), 109–118.

Sulisay Phonekeo, Tanvi Dave, Matthew Kern, Scott V Franklin, and David L Hu. 2016. Ant aggregations self-heal to compensate for the Ringelmann effect. *Soft Matter* 12, 18 (2016), 4214–4220.

Sulisay Phonekeo, Nathan Mlot, Daria Monaenkova, David L Hu, and Craig Tovey. 2017. Fire ants perpetually rebuild sinking towers. *Royal Society open science* 4, 7 (2017), 170475.

Stefan Popp and Anna Dornhaus. 2023. Ants combine systematic meandering and correlated random walks when searching for unknown resources. *Iscience* 26, 2 (2023).

Martin Pražák, Damien Maupu, Mungo Pay, Muhittin Bilginer, Aleksandar Atanasov, and Cristobal Infante Esquivel. 2018. Taming the swarm: rippers on pacific rim uprising. In *ACM SIGGRAPH 2018 Talks*. 1–2.

Daniel Ram, Theodore Gast, Chenfanfu Jiang, Craig Schroeder, Alexey Stomakhin, Joseph Teran, and Pirouz Kavehpour. 2015. A material point method for viscoelastic fluids, foams and sponges. In *Proceedings of the 14th ACM SIGGRAPH/Eurographics Symposium on Computer Animation*. 157–163.

Chris R Reid, Matthew J Lutz, Scott Powell, Albert B Kao, Iain D Couzin, and Simon Garnier. 2015. Army ants dynamically adjust living bridges in response to a cost–benefit trade-off. *Proceedings of the National Academy of Sciences* 112, 49 (2015), 15113–15118.

Jiaping Ren, Xinjie Wang, Xiaogang Jin, and Dinesh Manocha. 2016. Simulating flying insects using dynamics and data-driven noise modeling to generate diverse collective behaviors. *PloS one* 11, 5 (2016), e0155698.

Craig W Reynolds. 1987. Flocks, herds and schools: A distributed behavioral model. In *Proceedings of the 14th annual conference on Computer graphics and interactive techniques*. 25–34.

Michael Rubenstein, Alejandro Cornejo, and Radhika Nagpal. 2014. Programmable self-assembly in a thousand-robot swarm. *Science* 345, 6198 (2014), 795–799.

Daiki Satoi, Mikihiro Hagiwara, Akira Uemoto, Hisanao Nakadai, and Junichi Hoshino. 2016. Unified motion planner for fishes with various swimming styles. *ACM Transactions on Graphics (TOG)* 35, 4 (2016), 1–15.

Tetsuya Takahashi, Yoshinori Dobashi, Issei Fujishiro, and Tomoyuki Nishita. 2016. Volume preserving viscoelastic fluids with large deformations using position-based velocity corrections. *The Visual Computer* 32 (2016), 57–66.

Bilas Talukdar and Tomer Weiss. 2025. Generalized, Dynamic Multi-agent Torso Crowds. *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 8, 1 (2025), 1–15.

Jie Tan, Yuting Gu, Greg Turk, and C Karen Liu. 2011. Articulated swimming creatures. *ACM Transactions on Graphics (TOG)* 30, 4 (2011), 1–12.

Michael Tennenbaum, Zhongyang Liu, David Hu, and Alberto Fernandez-Nieves. 2016. Mechanics of fire ant aggregations. *Nature materials* 15, 1 (2016), 54–59.

Jur Van Den Berg, Stephen J Guy, Ming Lin, and Dinesh Manocha. 2011. Reciprocal n-body collision avoidance. In *Robotics research*. Springer, 3–19.

Wouter van Toll and Julien Pettré. 2021. Algorithms for Microscopic Crowd Simulation: Advancements in the 2010s. In *Computer Graphics Forum*, Vol. 40.

Franck J Vernerey, Tong Shen, Shankar Lalitha Sridhar, and Robert J Wagner. 2018. How do fire ants control the rheology of their aggregations? A statistical mechanics approach. *Journal of The Royal Society Interface* 15, 147 (2018), 20180642.

Robert J Wagner, Samuel C Lamont, Zachary T White, and Franck J Vernerey. 2024. Catch bond kinetics are instrumental to cohesion of fire ant rafts under load. *Proceedings of the National Academy of Sciences* 121, 17 (2024), e2314772121.

Xinjie Wang, Jiaping Ren, Xiaogang Jin, and Dinesh Manocha. 2015. BSwarm: biologically-plausible dynamics model of insect swarms. In *Proceedings of the 14th ACM SIGGRAPH/Eurographics Symposium on Computer Animation*. 111–118.

Tomer Weiss. 2023. Fast position-based multi-agent group dynamics. *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 6, 1 (2023), 1–15.

Tomer Weiss, Alan Litteneker, Chenfanfu Jiang, and Demetri Terzopoulos. 2019. Position-based real-time simulation of large crowds. *Computers & Graphics* 78 (2019), 12–22.

Wei Xiang, Jiaping Ren, Kuan Wang, Zhigang Deng, and Xiaogang Jin. 2019. Biologically inspired ant colony simulation. *Computer Animation and Virtual Worlds* 30, 5 (2019), e1867.

Wei Xiang, Xinran Yao, He Wang, and Xiaogang Jin. 2020. FASTSWARM: A data-driven framework for real-time flying insect swarm simulation. *Computer Animation and Virtual Worlds* 31, 4-5 (2020), e1957.

Yalan Zhang, Shen Long, Yanrui Xu, Xiaokun Wang, Chao Yao, Jiri Kosinka, Steffen Frey, Alexandru Telea, and Xiaojuan Ban. 2024. Multiphase Viscoelastic Non-Newtonian Fluid Simulation. In *Computer Graphics Forum*, Vol. 43. Wiley Online Library, e15180.